

Week 18; Course Synthesis and Assessment Preparation

Review; Reflect; Prepare

- Comprehensive review of course content and skills
- Identify and address any gaps in understanding
- Get ready for knowledge-based and project assessments

Learning Objectives

- Consolidate your knowledge of large-scale datasets and AI workflows
- Review tools, ethics, and processes used throughout the course
- Practice industry-relevant problem solving and project communication
- Prepare effectively for final assessments and workplace application

Week Agenda

1. Knowledge review; revisit key topics across all weeks
2. Practical activities and revision exercises
3. Assessment preparation; review requirements, examples, and tips
4. Planning for workplace readiness and next steps

Key Industry Context

- Focus on skills required in data engineering and AI research teams
- Practical use of open datasets and industry-standard tools
- Emphasize reproducibility, ethics, and responsible AI workflows
- Workplace context; AI labs and data-driven organizations

Industry Tools and Workflows

- HuggingFace Datasets for data operations and analysis
- Python tools; pandas, numpy, matplotlib, seaborn for EDA and visualization
- CLIP models for working with image and text data
- Jupyter Notebooks for transparent, reproducible projects

Review; Open Datasets in AI

- Examples; LAION-5B, COCO, UCI ML Repository datasets
- Processes; dataset collection, documentation, and access
- Challenges; data quality, representativity, openness
- Connects with Weeks 1, 2, 5, 6

Data Ethics and Compliance

- Key topics; bias, privacy, consent, compliance
- Standards; open dataset documentation, responsible data usage
- Legislative context; ensure your workflows follow best practices
- Connects with Weeks 3, 9, 16

Data Cleaning and Preprocessing

- Steps; handling missing data, normalization, deduplication
- Tools; pandas transform, groupby, fillna functions
- Example; Cleaning a UCI CSV dataset for analysis
- Connects with Weeks 4, 7, 8

Exploratory Data Analysis (EDA)

- Visual tools; matplotlib for plots, seaborn for trends
- Common EDA steps; describe statistics, visualize outliers, correlations
- Practice; Summarize results and key findings in notebooks
- Connects with Weeks 10, 11

Machine Learning Applications

- ML tasks; classification, regression, clustering on open datasets
- Workflow; data split, model training, evaluation, interpret results
- Practice; Use scikit-learn or PyTorch for simple models
- Connects with Weeks 13, 14, 15

Building Reproducible Data Pipelines

- Best practices; code versioning, data documentation, pipeline scripts
- Reproducibility; using notebooks and environment files
- Activity; Create a reproducible workflow for EDA or ML
- Connects with Weeks 8, 12

Practical Activity; Dataset Analysis Quick Review

- Choose a publicly available dataset; summarize its features and documentation
- List at least three strengths and two weaknesses of its structure or curation
- Identify ethical issues based on course frameworks
- Share your summary and feedback with peers

Assessment Preparation; Knowledge Test

- Format; written/online test on data analysis, ethics, documentation, critical thinking
- Tips; Focus on practical cases and show understanding of industry standards
- Revision; Revisit main readings and your past project reports
- Ask questions now if uncertain about any requirements

Assessment Preparation; Project Completion

- Checklist; Complete notebooks and reports for EDA and ML tasks
- Ensure all documentation standards are met
- Peer review; Exchange your work with a partner for feedback
- Submit final versions before deadline; check against assessment criteria

Industry Case Study; Real-World AI Data Pipeline

- Example; AI organization curating an open dataset for LLM training
- Steps followed; data collection, bias audit, documentation, release
- Challenges encountered; scale, privacy, evolving standards
- Reflection; How would you address similar challenges in your workplace?

Planning for Workplace Readiness

- Identify the most relevant technical and ethical skills for your career
- Practice communicating project findings and recommendations
- Explore opportunities to contribute to open data or AI projects
- Set goals for ongoing skill development after the course

Q&A; Identify Gaps and Next Steps

- Open discussion; What topics need more review?
- What questions do you have about assessments or workplace application?
- Plan any further revision or meetings as needed

Summary and Closing

- You are prepared to critically evaluate, analyse, and add value to AI datasets
- Apply your knowledge to assessments and in the workplace
- Thank you for your commitment; best of luck with final assessments and your career

